Pre-Analysis Plan (PAP)
Many Daughters Project: H1–H4

As of July 30, 2025

1 Overview

Dataset and unit. We use SOEP-Core individual panel data, merging `ppathl` (IDs), `biobirth` (fertility history), `pgen` (individual background), `pl` (individual questionnaire), and `hgen` (household income). The unit is individual–year (i, t). We declare the panel with `xtset pid syear` and cluster standard errors at `pid`.

Treatment. The treatment indicator `anydaughter_sofar` equals 1 if the respondent has at least one daughter *born on or before* survey year t. It is constructed from cleaned copies of `kidgeb01--kidgeb19` (birth years) and `kidsex01--kidsex19` (1=boy, 2=girl).

Controls (built exactly as in code).

- *Time-invariant (drop in individual FE):* `male` (`sex` = 1), `pgisced11` (ISCED-2011 education; negative codes → missing), `german` (`pgnation` = 1), `age1stb` (age at first birth), `sumkids` (total births).
- *Time-varying (must be included in FE):* `age` (`syear` − `gebjahr`), `working` (`pglfs` ∈ {11, 12}), `pglabnet` (negatives → missing; if missing and `working`=0 then set to 0), `marstat` (4-category recode of `pgfamstd`), `hghinc` (negatives → missing), `numkids_sofar`.

Common sample restrictions.

- Keep only *post-parenthood* waves: `numkids_sofar` > 0.
- Keep observations with valid outcomes and required controls; SOEP negative missings < 0 are recoded to Stata missing (`.`) before construction.

Model grid (reported for every hypothesis). Eight prespecified columns exported via `outreg2`:

1. OLS, no controls.
2. OLS + time-invariant controls.
3. OLS + all controls.
4. Individual FE (within), + time-varying controls.
5. OLS + year FE.
6. OLS + time-invariant controls + year FE.
7. OLS + all controls + year FE.
8. **Preferred:** two-way fixed effects (individual FE + year FE) + time-varying controls.

All columns use *cluster-robust* SE at `pid`.

Decision rule (common). Two-sided test of the treatment coefficient. If the estimated coefficient has the sign implied by the hypothesis and $p < 0.05$ (clustered at `pid`), we label the result *"in line with the hypothesis"*; otherwise *"not in line"*. For H4 we test the *interaction* coefficient (male–female difference).

2 H1: Daughters and Support for Gender Equality (Attitudes)

2.1 Hypothesis

H1. Having (at least one) daughter increases support for gender equality. Let X denote the attitudes index; we test $\mathbb{E}[X \mid \textit{anydaughter}] > \mathbb{E}[X \mid \textit{no daughter}]$.

2.2 Outcome (X): Gender-egalitarian attitudes index

From `pl`. We standardize items and average the z-scores so higher values mean more egalitarian attitudes.

- **Items (14):**
  - `plh0301` (Women should rather care about family than career).
  - `plh0302_v1` ($<$ 3 child suffers if mother works; 4-point).
  - `plh0302_v2` ($<$ 3 child suffers if mother works; 7-point; *reverse*).
  - `plh0303` (Men should be involved in housework; *reverse*).
  - `plh0304` (Child suffers when father focuses on career; *reverse*).
  - `plj0623` (Wife earns more $\Rightarrow$ problems; *reverse*).
  - `plj0624` (Educating sons more important; *reverse*).
  - `plj0625` (Husband should have final say; *reverse*).
  - `plh0395i01`–`i06` (gender identity / acceptance battery).
- **Cleaning & coding.** Convert SOEP negative missings to missing. Reverse-code items flagged above so that *higher = more egalitarian* for every item (4-point: $5 - x$; 7-point: $8 - x$). Z-score each adjusted item over the analysis sample, then compute X as the row mean over available z-items.

2.3 Treatment

`anydaughter_sofar` $= 1$ if $\exists$ daughter with `kidgeb#_c` $\leq$ `syear` and `kidsex#_c`$= 2$. Auxiliary counts: `numkids_sofar`, `numdaughters_sofar`.

2.4 Controls, sample, and missingness

As listed in Section 1. Income rule: if `pglabnet==.` and `working==0`, set `pglabnet=0`.

2.5 Specifications and target estimand

Let $W(i,t)$ denote the relevant control set. The preferred specification is:

$$X(i,t) = \beta\, \texttt{anydaughter_sofar}(i,t) + W(i,t)\gamma + \alpha_i + \gamma_t + \varepsilon(i,t),$$

with cluster-robust SE at `pid`. The target is β.

2.6 Inference and decision

Primary decision uses column 8 (two-way fixed effects). If $\hat{\beta} > 0$ and $p < 0.05$, we conclude "in line with H1".

2.7 Deliverables

One 8-column regression table (`h1_model_results.xls`); one CSV from column 8 (`h1_results.csv`) containing: model/method, coefficient, SE, p-value, N, interpretation.

2.8 Pre-specified robustness

(i) Replace `pgisced11` with `pgcasmin` or `pgbilzeit`; (ii) re-estimate excluding waves with sparse item coverage; (iii) compare OLS + year FE vs two-way fixed effects.

3 H2: Daughters and Left-Leaning Political Preferences

Hypothesis

H2. Having a daughter makes respondents more left-leaning.

3.1 Outcome (Y): Left–right self-placement (z-score)

From `plh0004` (0=far left, 10=far right). Reverse so higher = more left:

$$Y_{\text{raw}} = 10 - \texttt{plh0004}, \qquad Y = \frac{Y_{\text{raw}} - \overline{Y_{\text{raw}}}}{\text{sd}(Y_{\text{raw}})}.$$

SOEP negative missings are recoded to missing before construction.

3.2 Treatment, controls, sample

Same as H1 (including the `pglabnet` zeroing rule when not working) and post-parenthood restriction.

3.3 Specifications and target estimand

Preferred (two-way fixed effects):

$$Y(i,t) = \beta\, \texttt{anydaughter_sofar}(i,t) + W(i,t)\gamma + \alpha_i + \gamma_t + \varepsilon(i,t).$$

Target is β; cluster-robust SE at `pid`.

3.4 Inference, deliverables, robustness

Same decision rule as H1 (expect $\beta > 0$). Outputs: `h2_model_results.xls`, `h2_results.csv`. Robustness as in H1.

4 H3: Daughters and Within-Household Gender Equality (Behavior)

4.1 Hypothesis

H3. Having a daughter increases household gender equality in behavior/outcomes.

4.2 Outcome (Z): Housework equality index (0–1)

We merge partner information by building a compact partner file (from `pl`) with partner housework hours and labour-force status, renaming partner IDs to `pgpartnr`, and merging back on `pgpartnr` and `syear`.

Inputs (respondent & partner). From `pl`:

- Respondent: `pli0012_h` (Saturday), `pli0016_h` (Sunday), `pli0043_h` (weekday).
- Partner: corresponding variables renamed to `p_pli0012_h`, `p_pli0016_h`, `p_pli0043_h`.

Weighted daily hours. Using weights Sat = 1/7, Sun = 1/7, Weekday = 5/7, compute respondent `hw_daily` and partner `p_hw_daily`. Handle missing components by renormalizing the denominator when some components are missing.

Equality index.

$$Z = 1 - \frac{|\,\texttt{hw_daily} - \texttt{p_hw_daily}\,|}{\texttt{hw_daily} + \texttt{p_hw_daily}},$$

defined when both hours ≥ 0 and the denominator > 0. Interpretation: 1 = perfectly equal, 0 = maximally unequal.

4.3 Treatment, controls, sample

Treatment and controls as in H1. Additional sample requirement: partner variables must merge to compute Z. Post-parenthood restriction applies.

4.4 Specifications and target estimand

Preferred (two-way fixed effects):

$$Z(i,t) = \beta\, \texttt{anydaughter_sofar}(i,t) + W(i,t)\gamma + \alpha_i + \gamma_t + \varepsilon(i,t),$$

clustered at `pid`. Target is β.

4.5 Inference, deliverables, robustness

Decision: expect $\beta > 0$. Outputs: `h3_model_results.xls`, `h3_results.csv`. Robustness: (i) build Z using weekday hours only; (ii) cap extreme hours (e.g., 99th pct.); (iii) optionally add partner working dummy as a control (sensitivity check).

5 H4: Larger Attitude Effect for Men than for Women

5.1 Hypothesis

H4. The "daughter effect" on egalitarian attitudes is larger for men than for women:

$$\big(\Delta X \mid \text{male}\big) - \big(\Delta X \mid \text{female}\big) > 0.$$

5.2 Outcome, treatment, moderator

Outcome X is the H1 composite; treatment is `anydaughter_sofar`; moderator `male` = (`sex` = 1). Create the interaction

$$\texttt{anydaughter_sofar_male} = \texttt{anydaughter_sofar} \times \texttt{male}.$$

5.3 Specifications and parameters

Preferred (two-way fixed effects):

$$X(i,t) = \beta_1\, \texttt{anydaughter_sofar_male}(i,t) + \beta_2\, \texttt{anydaughter_sofar}(i,t) + W(i,t)\gamma + \alpha_i + \gamma_t + \varepsilon(i,t).$$

Interpretation: female effect = β_2; male effect = $\beta_2+\beta_1$; *difference (male−female)* = β_1 (target). Cluster SE at `pid`.

5.4 Inference, deliverables, robustness

Primary test: $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 > 0$. Decision "in line with H4" if $\hat{\beta}_1 > 0$ and $p < 0.05$. Outputs: `h4_model_results.xls`, `h4_results.csv`. Robustness as in H1 (stability of $\hat{\beta}_1$ across OLS+year FE and two-way fixed effects).